MARTA KOWAL-BRANDT

QA Automation Engineer | Test Infrastructure & DevOps

Professional Summary

QA automation and test-infrastructure engineer with hands-on experience designing GitLab CI pipelines, Kubernetes-based ephemeral environments, infrastructure observability, and a personal Terraform-managed VPS project. Built Kubernetes-based merge-request environments, improved pipeline speed and cost, and developed test observability dashboards. Personal infrastructure projects include a k3s cluster managed with GitOps and a Hetzner VPS provisioned with Terraform. AWS experience is limited to personal learning with EC2 and S3 basics.

Core Skills

CI/CD: GitLab CI, Jenkins, pipeline profiling and optimization, dependency caching, parallel test execution
Containers and orchestration: Docker, Kubernetes, k3s, Helm, Argo CD
Infrastructure as code and configuration: Terraform (personal project; VPS provisioning), Ansible (basic)
Monitoring and data: Prometheus, Grafana, Loki, PostgreSQL
Automation and testing: Java, Selenium, Python, pytest, Bash, pytest-xdist, end-to-end and regression testing
Cloud and Infrastructure Projects: AWS EC2 and S3 basics (personal learning); Hetzner VPS (personal project)

Professional Experience

QA Automation Engineer (Test Infrastructure) | Odrava Health Software *2021–Present*

- Own GitLab CI pipelines and ephemeral test environments for three teams.
- Designed and implemented the delivery and lifecycle workflow for merge-request environments used by QA and product teams for pre-merge reviews and end-to-end testing.
- Automated environment cleanup after merge or 48 hours of inactivity; enabled QA and product teams to review features before merge and run end-to-end tests against each environment.
- Profiled pipelines across three teams, parallelized test jobs, added dependency caching and separated end-to-end tests into a dedicated stage. Reduced average pipeline time from 38 to 14 minutes and monthly CI runner costs by about 30%.
- Collected CI test results in PostgreSQL and created Grafana dashboards to identify flaky tests by team. Flaky test rate fell from 7% to below 1% after teams addressed the leading offenders.
- Wrote the internal guide to how the CI system works; the document is used in onboarding.
- Mentored two QA interns in 2023.
- Work with namespace-level access on the shared Kubernetes cluster; the platform team operates the cluster.

QA Engineer | Brightmill Software House *2019–2021*

- Tested web and mobile applications for multiple clients.
- Introduced the first automated regression suite on the largest project, using Java and Selenium.
- Set up Jenkins to run regression tests nightly. Before automation, a full regression took two people three days.

Selected Infrastructure Projects

Homelab k3s Cluster with GitOps | Personal *2024–2025*

- Built a three-node Raspberry Pi 5 k3s cluster, with applications deployed from a Git repository using Argo CD.

- Configured Traefik ingress and cert-manager for TLS certificate management; added Prometheus, Grafana, and Loki for monitoring.
- Provisioned infrastructure for a small Hetzner VPS with Terraform.
- Technologies: k3s, Argo CD, Prometheus, Grafana, Loki, Terraform, Ansible (basic).

Merge-Request Ephemeral Environments | Odrava Health *2023–2024*

- Implemented short-lived merge-request environments for the application, PostgreSQL, and seeded data on a shared Kubernetes cluster, with automated lifecycle management.
- Technologies: GitLab CI, Helm, Kubernetes, Docker, Bash, PostgreSQL.

Flaky Test Dashboard | Odrava Health *2023*

- Collected CI test results in PostgreSQL and built Grafana dashboards showing flaky tests by team.
- Flaky test rate fell from 7% to below 1% after teams began fixing the most frequent offenders.
- Technologies: Python, PostgreSQL, Grafana, GitLab API.

CI Pipeline Speed-up | Odrava Health *2022*

- Optimized pipelines across three teams through parallelized test jobs, dependency caching, and a dedicated end-to-end test stage.
- Technologies: GitLab CI, Docker, pytest-xdist.

Certifications & Professional Development

- ISTQB Foundation Level, 2019
- Certified Kubernetes Administrator (CKA): studying; exam booked for January 2027 (not yet passed)
- Terraform Associate: completed practice tests; certification exam not taken